

INFINITESIMAL MODEL PERTURBATION INFLUENCE IN THE LINEAR MIXED MODEL

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Summary: The usual methods of fitting the linear mixed model assume that there are no influential observations in the data. However, the presence of influential observations in the dataset can completely distort an analysis and lead to inappropriate actions and conclusions. We present a model perturbation approach to identify influential observations. All the diagnostic tools are a function of leverages and outliers. Moreover, a particularly simple special case of the model perturbation approach to case deletions is given to circumvent some of the criticisms about the suitability of case deletion diagnostics in the mixed models. The method is illustrated with an example. The method seems to provide a relatively simple and unified approach for diagnostics.

1. Introduction

The linear mixed model has been used widely in the biological, agricultural, behavioural, physical and engineering sciences. It is well known that the variables, the observations and the model assumptions fully determine the model. As an integral part of the model building, it is essential to build confidence and have reassurance that what we have fitted is not contradicted or unduly influenced by particular observations and/or assumptions. However, data analysts generally fit a mixed model to data without assessing the overall validity of the model itself (Oman, 1995). The basic reasons are (1) the lack

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of routine methods (in the literature and standard computing packages) for performing such analysis, and (2) the presumably high costs (for an analyst and computer time) in doing so.

Diagnostic methods are useful for assessing the adequacy of the assumptions underlying a modelling process, and for identifying unexpected characteristics of the data that may seriously influence conclusions or require special attention. Based on the empirical influence function of the regression coefficient, Cook (1977) introduced a statistic to investigate the influence of an observation on the regression fittings. Since then many measures of influence have been proposed for identifying a subset of the observations as outliers or influential.

Cook and Weisberg (1982) described influence as the change in the outcome of the analysis as a result of small perturbations in the problem formulation. There have been two main approaches to assessing the influence of a single observation. The first approach measures influence by the difference in point predictions from the full and the reduced data, scaled by the estimate of the standard deviation of the prediction (Cook and Weisberg, 1982, page 126 and Belsley, Kuh and Welsch 1980, page 15). The second approach is based on the perturbation of a case and not on its total deletion, and employs a differential comparison of point predictions before and after perturbation (Pregibon, 1981, Belsley et al., 1980, page 24 and Lawrance, 1990).

Although the assessment of the influence using the model perturbation approach is generally considered to be useful in ordinary regression, as far as we are aware no extensions are available for carrying out such assessment in the context of the linear mixed model. Much of the recent work is

concerned with the extension of the case deletions approach. Among others, Christensen, Pearson and Johnston (1992), Banerjee and Frees (1997), Haslett and Dillane (2004) and Zewotir and Galpin (2005) extended a linear regression case deletion diagnostics approach to the linear mixed model.

Although case deletion is a well-established approach, there are some pitfalls in its suitability as a mixed model diagnostic. The main criticism is that deletion of data from any balanced design almost always destroys the balance and hence computational difficulties arise even for estimates of the fixed effect parameters. Haslett and Dillane (2004) proposed the “delete=replace” approach as an alternative approximate procedure to lighten the computational burden. The other criticism of the case deletion approach is that it requires, after an observation has been deleted, that each level of the fixed effects should have at least one observation. In other words, each level of the fixed effects factor should originally have at least two replications for the case deletion diagnostics to be applicable; otherwise, case deletion causes a change in the estimability of the fixed effects, through the reduction of the design matrix rank by one after deletion.

Besides the case deletion approach, other approaches also exist to assess the influence of an observation in a linear model context. Beckman, Nachtsheim and Cook (1987) developed a local influence diagnostics approach suggested by Cook (1986). However, since the construction of maximal normal curvature and the direction cosines involves an eigenvalue analysis, the approach is computationally expensive. The method becomes inappropriate whenever any first derivative of the log-likelihood evaluated at the MLE is not zero. This situation may happen if the variance components estimators are constrained to be non-negative (which is, in fact, usually the case). Moreover, since there is usually no simple analytical expression for maximal normal curvature and

the direction cosines, it is very difficult to attribute the source of influence to interpretable components such as residuals and leverages. Lesaffre and Verbeke (1998) proposed the case weight perturbation approach to assess the influential subjects in linear mixed models for longitudinal data. They considered change in the parameter estimates due to the change in the weights of the log-likelihood contributions of specific individuals. Lesaffre and Verbeke (1998) considered a different problem (subject influence, through repeated observations) to the one considered here (case influence).

Infinitesimal influence can be the influence of small changes in the individual observations (called infinitesimal data influence) or the influence of small (infinitesimal) perturbation of the model (called infinitesimal model influence). Demidenko (2004, Chapter 9) provided a comprehensive introduction to infinitesimal data influence diagnostics on linear regression and logistic regression. He further extended the infinitesimal data influence diagnostics approach to a linear mixed model (Demidenko, 2004 Section 9.9, p. 522-527). However, the approach we consider in this paper is an infinitesimal model influence diagnostic for linear mixed models.

Instead of deleting cases, the problem of local sensitivity analysis of the mixed model is examined here. In particular, we investigate how the estimates (or predictors) change if the constancy of error variances is changed infinitesimally. A particular parallel between case deletion and such a perturbation approach is that case deletion can be regarded as an infinite perturbation to the variance of a single case. This approach solves the criticisms applied to the case deletion approach in the mixed model.

We illustrate the application of the model perturbation method with an example. The data in this example has been used in similar studies (Beckman et al., 1987, Christensen et al., 1992, Haslett and Dillane, 2004 and Zewotir

and Galpin, 2005). The example shows that the approaches presented in this paper may be useful in practice.

2. The model

We consider the linear mixed model

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \varepsilon,$$

where β is a $p \times 1$ vector of unknown constants, the fixed effects of the model; $\mathbf{X}$ is an $n \times p$ known design matrix of fixed numbers associated with β ; $\mathbf{Z} = [\mathbf{Z}_1 \mid \mathbf{Z}_2 \mid \dots \mid \mathbf{Z}_r]$, where $\mathbf{Z}_i$ is an $n \times q_i$ known design matrix of the random effect factor i ; $\mathbf{u}' = [\mathbf{u}'_1 \mid \mathbf{u}'_2 \mid \dots \mid \mathbf{u}'_r]$, where $\mathbf{u}_i$ is a $q_i \times 1$ vector of independent random variables from $N(0, \sigma_i^2)$, $i = 1, 2, \dots, r$; ε is an $n \times 1$ vector of independent error terms from $N(0, \sigma_e^2)$; and $\mathbf{u}_i$ and ε are mutually independent. One may also write $\mathbf{u} \sim N(\mathbf{0}, \sigma_e^2 \mathbf{D})$, where $\mathbf{D}$ is block diagonal with the i^{th} block being $\gamma_i \mathbf{I}_{q_i}$, for $\gamma_i = \sigma_i^2 / \sigma_e^2$, so that $\mathbf{y}$ has a multivariate normal distribution with $E(\mathbf{y}) = \mathbf{X}\beta$ and $\text{var}(\mathbf{y}) = \sigma_e^2 \mathbf{H}$, for $\mathbf{H} = \mathbf{I}_n + \mathbf{Z}\mathbf{D}\mathbf{Z}' = \mathbf{I}_n + \sum_{i=1}^r \gamma_i \mathbf{Z}_i \mathbf{Z}_i'$.

If the γ_i 's (and hence $\mathbf{H}$) are known, the maximum likelihood estimates of β , σ_e^2 , and the realized value of $\mathbf{u}$ are given by $\hat{\beta} = (\mathbf{X}'\mathbf{H}^{-1}\mathbf{X})^{-1} \mathbf{X}'\mathbf{H}^{-1}\mathbf{y}$, $\hat{\sigma}_e^2 = (\mathbf{y} - \mathbf{X}\hat{\beta})' \mathbf{H}^{-1} (\mathbf{y} - \mathbf{X}\hat{\beta}) / n$, $\tilde{\mathbf{u}} = \mathbf{D}\mathbf{Z}'\mathbf{H}^{-1} (\mathbf{y} - \mathbf{X}\hat{\beta})$ respectively. Here $\hat{\beta}$ is the best linear unbiased estimator (BLUE) of β and $\tilde{\mathbf{u}}$ is the best linear unbiased predictor (BLUP) of $\mathbf{u}$. If the γ_i 's are unknown, the maximum likelihood estimates $\hat{\gamma}_i$ are substituted back into $\mathbf{H}$ to obtain $\hat{\beta}$, $\hat{\sigma}_e^2$ and $\tilde{\mathbf{u}}$ (see, e.g., Harville, 1977, Searle, Casella and McCulloch, 1992, SAS Institute, 1992, McCulloch and Searle, 2001). Note that even if our discussion throughout the paper focuses on maximum likelihood estimation, it applies to restricted maximum likelihood estimation and minimum norm quadratic unbiased estimation.

The definition of residuals in the fixed effects model is $\mathbf{y} - \mathbf{X}\hat{\beta}$ whereas in the mixed model it is $\hat{\mathbf{y}} - \mathbf{X}\hat{\beta} - \mathbf{Z}\tilde{\mathbf{u}}$ (for more discussions see Zewotir and Galpin, 2007). The fitted values of the response variable $\mathbf{y}$ are $\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} + \mathbf{Z}\tilde{\mathbf{u}}$, giving the residual vector $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$. The symmetric matrix $\mathbf{R} = \mathbf{H}^{-1} - \mathbf{H}^{-1}\mathbf{X}(\mathbf{X}\mathbf{H}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{H}^{-1}$ transforms the observations into residuals, that is, $\mathbf{e} = \mathbf{R}\mathbf{y}$ and $\text{var}(\mathbf{e}) = \mathbf{R}\mathbf{H}\mathbf{R} = \mathbf{R}$. The correlation between e_i and e_j is entirely determined by the elements of $\mathbf{R}$, $\text{corr}(e_i, e_j) = r_{ij}/\sqrt{r_{ii}r_{jj}}$. The magnitude of the diagonal elements of $\mathbf{R}$ ranges from 0 to 1. If r_{ii} approaches 0 or 1 then r_{ij} approaches 0 for $i \neq j$ (Zewotir and Galpin, 2007). The i^{th} Studentized residual is defined as $t_i = e_i/\sqrt{\text{var}(e_i)} = e_i/(\hat{\sigma}_e\sqrt{r_{ii}})$.

3. Perturbations

Suppose the perturbation is ω ; it is a mathematical quantity introduced into the model, not a parameter to be estimated and not data. The estimate (or predictor) is then obtained as a function of ω , say $\hat{\theta}(\omega)$. Let the null perturbation be ω_0 , with $\hat{\theta}(\omega_0)$ being the estimate (or predictor) for the unperturbed model. Unlike the global influence assessment which involves the comparison of a general $\hat{\theta}(\omega)$ and $\hat{\theta}(\omega_0)$ our assessment focuses on the comparison of $\hat{\theta}(\omega)$ when ω departs just slightly from ω_0 . This contrast leads to increased mathematical tractability via the use of derivatives with respect to ω .

The perturbation approach is obtained by specifying $\varepsilon_i \sim N(0, \sigma_e^2/\omega)$, where $0 < \omega < \infty$ and ε_i is the i^{th} element of ε in the specified linear mixed model $\mathbf{y} = \mathbf{X}\beta + \sum_{i=1}^r \mathbf{Z}_i\mathbf{u}_i + \varepsilon$. The perturbed $\mathbf{H}$ is then denoted by $\mathbf{H}_\omega$,

$$\mathbf{H}_\omega = \mathbf{H} - (1 - 1/\omega) \mathbf{ii}'$$

where $\mathbf{i}$ is the i^{th} column of the identity matrix. Without loss of generality, for simplicity, we partition the matrices as if the i^{th} perturbed observation is the first observation; i.e. $i = 1$. Then

$$\mathbf{C} = \mathbf{H}^{-1} = \begin{bmatrix} c_{ii} & \mathbf{c}'_{(i)} \\ \mathbf{c}_{(i)} & \mathbf{C}_{[i]} \end{bmatrix} \quad \text{and} \quad \mathbf{c}_i = \begin{bmatrix} c_{ii} \\ \mathbf{c}_{(i)} \end{bmatrix}.$$

Likewise $\mathbf{R}$ is partitioned as

$$\mathbf{R} = \begin{bmatrix} r_{ii} & \mathbf{r}'_{(i)} \\ \mathbf{r}_{(i)} & \mathbf{R}_{[i]} \end{bmatrix} \quad \text{and} \quad \mathbf{r}_i = \begin{bmatrix} r_{ii} \\ \mathbf{r}_{(i)} \end{bmatrix}.$$

Noting $r_{ii} = c_{ii} - \mathbf{c}'_i \mathbf{X} \mathbf{W} (\mathbf{X}, \mathbf{X})^{-1} \mathbf{X}' \mathbf{c}_i$ it follows that $c_{ii} \geq r_{ii}$. From the matrix identity

$$(\mathbf{G} - a \mathbf{t} \mathbf{t}')^{-1} = \mathbf{G}^{-1} + a \mathbf{G}^{-1} \mathbf{t} \mathbf{t}' \mathbf{G}^{-1} / (1 - a \mathbf{t}' \mathbf{G}^{-1} \mathbf{t})$$

it follows that

$$\begin{aligned} \mathbf{H}_\omega^{-1} &= \mathbf{C} - \frac{(\omega - 1) \mathbf{c}_i \mathbf{c}'_i}{c_{ii} (\omega - 1) - \omega}; \\ (\mathbf{X}' \mathbf{H}_\omega^{-1} \mathbf{X})^{-1} &= \left(\mathbf{X}' \mathbf{C} \mathbf{X} - \frac{(\omega - 1)}{c_{ii} (\omega - 1) - \omega} \mathbf{X}' \mathbf{c}_i \mathbf{c}'_i \mathbf{X} \right)^{-1} \\ &= (\mathbf{X}' \mathbf{C} \mathbf{X})^{-1} + \frac{(\omega - 1) (\mathbf{X}' \mathbf{C} \mathbf{X})^{-1} \mathbf{X}' \mathbf{c}_i \mathbf{c}'_i \mathbf{X} (\mathbf{X}' \mathbf{C} \mathbf{X})^{-1}}{r_{ii} (\omega - 1) - \omega} \end{aligned}$$

and that

$$\mathbf{R}_\omega = \mathbf{H}_\omega^{-1} - \mathbf{H}_\omega^{-1} \mathbf{X} (\mathbf{X}' \mathbf{H}_\omega^{-1} \mathbf{X})^{-1} \mathbf{X}' \mathbf{H}_\omega^{-1} = \mathbf{R} - \frac{(\omega - 1) \mathbf{r}_i \mathbf{r}'_i}{r_{ii} (\omega - 1) - \omega}.$$

The fixed effect parameter estimate, as a function of the perturbation, say $\hat{\beta}(\omega)$, is then

$$\hat{\beta}(\omega) = \hat{\beta} - \frac{(\omega - 1) (\mathbf{X} \mathbf{C} \mathbf{X})^{-1} \mathbf{X}' \mathbf{c}_i e_i}{r_{ii} (\omega - 1) - \omega}.$$

The effect of infinitesimal perturbations of the variance of the i^{th} data point error term on the fixed effect parameter estimate is easily obtained by differentiation of $\hat{\beta}(\omega)$. This operation leads to

$$\Delta \hat{\beta}(\omega) = \frac{\partial \hat{\beta}(\omega)}{\partial \omega} = \frac{(\mathbf{X} \mathbf{C} \mathbf{X})^{-1} \mathbf{X}' \mathbf{c}_i e_i}{\{(\omega - 1) r_{ii} - \omega\}^2}.$$

Large values of this derivative indicate observations that have a large influence on the MLE of β . For diagnostic purposes, $\Delta\hat{\beta}(\omega)$ is sensitive to both outlying responses and extreme design points. Note that small values of r_{ii} indicate extreme design (high leverage) points.

The random effect predictor, as a function of the perturbations, is

$$\tilde{\mathbf{u}}(\omega) = \tilde{\mathbf{u}} - \frac{(\omega - 1) \mathbf{DZ}' \mathbf{r}_i e_i}{r_{ii} (\omega - 1) - \omega}.$$

The effect of infinitesimal perturbation of the error term variance of the i^{th} observation on the BLUP of the random effects is

$$\Delta\tilde{\mathbf{u}}(\omega) = \frac{\partial\tilde{\mathbf{u}}(\omega)}{\partial\omega} = \frac{\mathbf{DZ}' \mathbf{r}_i e_i}{\{(\omega - 1) r_{ii} - \omega\}^2}.$$

Clearly $\Delta\tilde{\mathbf{u}}(\omega)$ is sensitive to both outlying response values and high leverage points, at the i^{th} datum.

The MLE of the error term variance as a function of the perturbations, which accommodate the two limiting cases $\omega = 1$ and 0 as unperturbed and case deletion estimates, respectively, is

$$\hat{\sigma}_e^2(\omega) = \frac{n\hat{\sigma}_e^2}{n - (1 - \omega)} - \frac{(\omega - 1) e_i^2}{\{n - (1 - \omega)\} \{r_{ii} (\omega - 1) - \omega\}}.$$

An alternative perturbation would be $\varepsilon_i \sim N(0, (1 + \nu)\sigma_e^2)$ where $-1 \leq \nu < \infty$. In this case $\mathbf{H}_\nu = \mathbf{H} + \nu\mathbf{i}\mathbf{i}'$, and at $\nu = 0$, it is a null perturbation. Since $\nu = 1/\omega - 1$ this perturbation eventually leads to the same conclusion.

4. Measure of influence

The quantities introduced in the previous section should indicate which (if any) of the single observations one at a time (1) are not well explained by the model, or (2) are dominating some aspects of the fit. Since Cook's distance, Cook (1977), has become so well established in deletion influence, it seems worthwhile to stress on analogized influence measure. One general way of

measuring influences is by their likelihood discrepancy; and thus we use the log-likelihood displacement (or equivalently Akaike's Information Criterion (AIC), Akaike, 1974) as a measure of influence. The quantities, however, cannot exhaustively measure the effect on all components of the fitted model.

4.1 Log-likelihood displacement

We propose that the influence of the model perturbation be measured by the log likelihood (or AIC). If -2 times the log likelihood is an extremely large number all the subsequent results of the mixed model should be regarded with suspicion (SAS Institute, 1992 and references therein). We thus assess the change in -2 times the log likelihood.

The -2 times the log-likelihood function evaluated for the MLEs of β and σ_e^2 , after perturbation, becomes

$$\begin{aligned} \ell(\omega) = & n \log(2\pi) + n \log \hat{\sigma}_e^2(\omega) + \log(\det(\mathbf{H}_\omega)) \\ & + \frac{(\mathbf{y} - \mathbf{X}\hat{\beta}(\omega))' \mathbf{H}_\omega^{-1} (\mathbf{y} - \mathbf{X}\hat{\beta}(\omega))}{\hat{\sigma}_e^2(\omega)}. \end{aligned}$$

The change in $\ell(\omega)$ is then

$$\begin{aligned} \Delta\ell(\omega) = \frac{\partial\ell(\omega)}{\partial\omega} = & \frac{t_i^2 r_{ii} \frac{n + (r_{ii} - 1)(\omega - 1)}{n \{r_{ii}(\omega - 1) - \omega\}^2} - 1}{\{n - (1 - \omega)\} \left(1 - \frac{t_i^2 r_{ii}(\omega - 1)}{n \{r_{ii}(\omega - 1) - \omega\}}\right)} \\ & + \frac{c_{ii}}{\omega \{c_{ii}(\omega - 1) - \omega\}}. \end{aligned}$$

The 'likelihood distance', is then

$$LD_\omega = |\Delta\ell(\omega)|.$$

The larger the statistic LD_ω the higher the influence of the i^{th} observation perturbation on the fitted values.

4.2 Analogue of Cook's distance for fixed effects

One of the most important influential measurements in the ordinary regression analysis is Cook's distance (Cook, 1977, Cook and Weisberg, 1982), which can be used for assessing the influence of perturbations on the regression coefficients. For fixed effects estimates in the linear mixed model, Cook's distance can be defined as

$$\begin{aligned} CD_{\omega}(\beta) &= \frac{(\Delta \hat{\beta}(\omega))' (\mathbf{X}' \mathbf{C} \mathbf{X}) (\Delta \hat{\beta}(\omega))}{p \hat{\sigma}_e^2} \\ &= \frac{r_{ii}^2 t_i^2 \left(\frac{c_{ii}}{r_{ii}} - 1 \right)}{p [(\omega - 1) r_{ii} - \omega]^4}. \end{aligned}$$

A large value of $CD_{\omega}(\beta)$ indicates that the i^{th} observation is influential on $\hat{\beta}$.

4.3 Analogue of Cook's distance for random effects

The proposed diagnostic measure examines the change in the predictor of the random effects relative to the variance of the random effects, $var(\mathbf{u}) = \sigma_e^2 \mathbf{D}$.

$$\begin{aligned} CD_{\omega}(\mathbf{u}) &= (\Delta \tilde{\mathbf{u}}(\omega))' \mathbf{D}^{-1} (\Delta \tilde{\mathbf{u}}(\omega)) / \hat{\sigma}_e^2 \\ &= e_i^2 \mathbf{r}_i' \mathbf{Z} \mathbf{D} \mathbf{Z}' \mathbf{r}_i / \left[\hat{\sigma}_e^2 \{(\omega - 1) r_{ii} - \omega\}^4 \right] \\ &= e_i^2 (r_{ii} - \mathbf{r}_i' \mathbf{r}_i) / \left[\hat{\sigma}_e^2 \{(\omega - 1) r_{ii} - \omega\}^4 \right] \\ &= r_{ii}^2 t_i^2 \left(1 - \frac{ssq(\mathbf{r}_i)}{r_{ii}} \right) / \{(\omega - 1) r_{ii} - \omega\}^4 \end{aligned}$$

where $ssq(\mathbf{b})$ denotes the sum of the squares of the elements of $\mathbf{b}$. Note that $r_{ii} \geq ssq(\mathbf{r}_i)$ follows from $\mathbf{Z} \mathbf{D} \mathbf{Z}'$ is a positive semi definite matrix and $\mathbf{r}_i' \mathbf{Z} \mathbf{D} \mathbf{Z}' \mathbf{r}_i = r_{ii} - ssq(\mathbf{r}_i)$. A large $CD_{\omega}(\mathbf{u})$ indicates that the i^{th} observation is influential on predicting the random effects.

4.4 Influence on variance ratios

In all the above discussions we assumed that the variance ratios γ_i are known, or the estimates of the variance ratios are correct. If either of these assumptions is dubious, using the approach of Pregibon (1981) “one-step” diagnostic for variance ratios is essential. In this situation, we first examine the variance ratio influential points diagnostic and take any necessary remedial action to influential cases, if any, so that the above influence diagnostics would be performed with the confidence that the specified covariance matrix $\mathbf{D}$ is equally sensitive to each observation under the perturbation. In fact, since the fixed effects estimate and the variance components estimate are asymptotically independent, if the prime interest of analysis is on the fixed effects, an assessment of variance components might not be that mandatory. Even for a sample size of 36 ($n = 36$), Zewotir and Galpin (2005) found that variance ratios influential observations have no remarkable effect on the influential points of the fixed effects estimate.

A Newton-Raphson (NR) and/or Fisher scoring algorithm, which require first and second order partial derivatives, may be applied to obtain the maximum likelihood estimate of $\gamma' = [\gamma_1 \dots \gamma_r]$. For the NR iterative procedure, $\gamma^{(t+1)} = \gamma^{(t)} - (\mathbf{F}^{(t)})^{-1} \mathbf{g}^{(t)}$, while for the Fisher scoring algorithm, $\gamma^{(t+1)} = \gamma^{(t)} - (E)(\mathbf{F}^{(t)})^{-1} \mathbf{g}^{(t)}$, where, $\mathbf{g}^{(t)}$ and $\mathbf{F}^{(t)}$ are the gradient (first derivative of the log-likelihood) and the Hessian (second derivative of the log-likelihood) evaluated at $\gamma^{(t)}$, respectively. The i^{th} element of the gradient vector $\mathbf{g}$ is $-\frac{1}{2} \text{trace}(\mathbf{Z}_i \mathbf{C} \mathbf{Z}_i) + \frac{1}{2\sigma_e^2} \mathbf{d}' \mathbf{C} \mathbf{Z}_i \mathbf{Z}_i' \mathbf{C} \mathbf{d}$; the $(i, j)^{\text{th}}$, $i, j = 1, 2, \dots, r$, element of $\mathbf{F}$ is $\frac{1}{2} \text{trace}(\mathbf{Z}_i' \mathbf{C} \mathbf{Z}_j \mathbf{Z}_j' \mathbf{C} \mathbf{Z}_i) - \frac{1}{\sigma_e^2} \mathbf{d}' \mathbf{C} \mathbf{Z}_i \mathbf{Z}_i' \mathbf{C} \mathbf{Z}_j \mathbf{Z}_j' \mathbf{C} \mathbf{d}$ implying that the expected value of the $(i, j)^{\text{th}}$ element of $\mathbf{F}$ is $-\frac{1}{2} \text{trace}(\mathbf{Z}_i' \mathbf{C} \mathbf{Z}_j \mathbf{Z}_j' \mathbf{C} \mathbf{Z}_i)$.

If estimation is hampered by the iterative computational burden, a one-step estimator can be used as a way out (see, for example, Christensen et al., 1992, Fan and Chen, 1999, Haslett and Dillane, 2004, and Zewotir and Galpin, 2005). The use of one-step methods to create diagnostics for models involving nonlinear parameters was pioneered by Pregibon (1981). In general, such one-step diagnostics adequately estimate the fully iterated change in the parameter as long as the initial estimators in the iterative processes are consistent (Cox and Hinkley, 1974). The fact that we use the estimates from the unperturbed data set as the initial values, and the fact that these initial estimators are consistent, gives us the theory necessary to consider the one-step estimate as accurate enough to substitute the full step estimates for the diagnostic purposes.

Noting that at convergence $\mathbf{F} = E(\mathbf{F})$, the one-step updating formula for variance ratios becomes

$$\hat{\gamma}_{(\omega)} = \hat{\gamma} + (\mathbf{I} + \mathbf{V}\mathbf{G})^{-1} \mathbf{V}\mathbf{g}_{(\omega)}$$

where

$$\mathbf{V} = \text{var}(\hat{\gamma}) = -[E(\mathbf{F})]^{-1}; \quad \mathbf{G}(j, k) = \frac{1}{2}(\omega - 1)^2 \text{ssq}(\mathbf{Z}'_j \mathbf{c}_i \mathbf{c}'_i \mathbf{Z}_k) / \{c_{ii}(\omega - 1) - \omega\}^2 - (\omega - 1) \text{trace}(\mathbf{Z}'_j \mathbf{C} \mathbf{Z}_k \mathbf{Z}'_k \mathbf{c}_i \mathbf{c}'_i \mathbf{Z}_j) / \{c_{ii}(\omega - 1) - \omega\};$$

the j^{th} element of $\mathbf{g}_{(\omega)}$ is

$$\frac{\partial \gamma(\omega)}{\partial \gamma_{j(\omega)}} = \frac{\partial \gamma}{\partial \gamma_j} + \frac{\theta_{ji}}{\hat{\sigma}_{e(\omega)}^2} + \frac{1}{2}(\omega - 1) \text{ssq}(\mathbf{c}'_i \mathbf{Z}_j) / \{c_{ii}(\omega - 1) - \omega\},$$

for

$$\begin{aligned} \theta_{ji} = & \frac{1}{2} \text{ssq}(\mathbf{Z}'_j \mathbf{e}) \left[1 - \frac{n}{n - (1 - \omega)} + \frac{t_i^2 r_{ii}(\omega - 1)}{\{n - (1 - \omega)\} \{r_{ii}(\omega - 1) - \omega\}} \right] \\ & + \frac{1}{2} \text{ssq}(\mathbf{Z}'_j \mathbf{r}_i) \frac{e_i^2(\omega - 1)^2}{\{r_{ii}(\omega - 1) - \omega\}^2} - \mathbf{e}' \mathbf{Z}_j \mathbf{Z}'_j \mathbf{r}_i \frac{e_i(\omega - 1)}{\{r_{ii}(\omega - 1) - \omega\}} \end{aligned}$$

and $j, k = 1, 2, \dots, r$. But at convergence $\partial\gamma/\partial\gamma_j \rightarrow 0$, and hence the j^{th} element of $\mathbf{g}_{(\omega)}$ is

$$\frac{\theta_{ji}}{\hat{\sigma}_{e(\omega)}^2} + \frac{1}{2} (\omega - 1) ssq(\mathbf{c}'_i \mathbf{Z}_j) / \{c_{ii}(\omega - 1) - \omega\}.$$

Since $\mathbf{Z}_j$ is an $n \times q_j$ known design matrix of the random effect factor j , $\mathbf{Z}'_j \mathbf{e}$ is a $q_j \times 1$ vector of the sum of residuals corresponding to each of the q_j levels of random factor j . Likewise, the $q_j \times 1$ vectors $\mathbf{Z}'_j \mathbf{r}_i$ and $\mathbf{Z}'_j \mathbf{c}_i$ are the sum of the elements of $\mathbf{r}_i$ and $\mathbf{c}_i$ matched to each of the q_j levels of random factor j , respectively (for more discussions see Zewotir and Galpin, 2007).

Clearly, the one-step estimate can only be obtained by explicitly recomputing and reinverting $(\mathbf{I} + \mathbf{V}\mathbf{G})^{-1}$ for each i . The matrix $\mathbf{V} = var(\hat{\gamma})$ is available in the full model and there is no need to compute it for each i . At each step, $\mathbf{G}$ has to be computed. This matrix is an $r \times r$ matrix, where r is the number of random effects in the model. In most practical cases r will not be large enough to imply that an $r \times r$ matrix, $(\mathbf{I} + \mathbf{V}\mathbf{G})$, inversion is expensive.

The effect of infinitesimal perturbation of the error term variance of the i^{th} observation on the variance ratios estimate becomes

$$\Delta \hat{\gamma}(\omega) = \frac{\partial \tilde{\mathbf{u}}(\omega)}{\partial \omega} = -(\mathbf{I} + \mathbf{V}\mathbf{G})^{-1} \mathbf{V} \left\{ \mathbf{b} + \mathbf{A}(\mathbf{I} + \mathbf{V}\mathbf{G})^{-1} \mathbf{V} \mathbf{g}_\omega \right\}$$

where $\mathbf{A}(j, k) = (\omega - 1) ssq(\mathbf{Z}'_j \mathbf{c}_i \mathbf{c}'_i \mathbf{Z}_k) / \{(\omega - 1) c_{ii} - \omega\}^3 - \mathbf{c}'_i \mathbf{Z}_j \mathbf{Z}'_j \mathbf{C} \mathbf{Z}_k \mathbf{Z}'_k \mathbf{c}_i / \{(\omega - 1) c_{ii} - \omega\}^2$ for $j, k = 1, 2, \dots, r$; and the j^{th} element of $\mathbf{b}$ is

$$\mathbf{b}_j = \frac{\hat{\sigma}_{e(\omega)}^2 \phi_{ji} + \theta_{ji} \frac{n \hat{\sigma}_e^2}{\{n - (1 - \omega)\}^2} \left[\frac{t_i^2 r_{ii} \{n + (r_{ii} - 1)(\omega - 1)\}}{n \{r_{ii}(\omega - 1) - \omega\}^2} \right]}{\hat{\sigma}_{e(\omega)}^4} + \frac{1}{2} ssq(\mathbf{c}'_i \mathbf{Z}_j) / \{c_{ii}(\omega - 1) - \omega\}^2$$

for

$$\begin{aligned} \phi_{ji} = & \left[\frac{1}{2} ssq(\mathbf{Z}_j' \mathbf{e}) \frac{n - t_i^2 r_{ii} \{n + (r_{ii} - 1)(\omega - 1)\}}{\{n - (1 - \omega)\}^2} \right. \\ & \left. + ssq(\mathbf{Z}_j' \mathbf{r}_i) \frac{(\omega - 1) e_i^2}{\{r_{ii}(\omega - 1) - \omega\}} - e_i \mathbf{e}' \mathbf{Z}_j \mathbf{Z}_j' \mathbf{r}_i \right] \\ & / \{r_{ii}(\omega - 1) - \omega\}^2. \end{aligned}$$

The analogue of Cook's distance measure for variance ratios, is

$$\begin{aligned} CD_{\omega}(\gamma) &= \left(\Delta \hat{\gamma}_{(\omega)} \right)' \{var(\hat{\gamma})\}^{-1} \left(\Delta \hat{\gamma}_{(\omega)} \right) \\ &= \mathbf{k}' \mathbf{V}^{-1} \mathbf{k} \end{aligned}$$

where $\mathbf{k} = (\mathbf{I} + \mathbf{V}\mathbf{G})^{-1} \mathbf{V} \left\{ \mathbf{b} + \mathbf{A}(\mathbf{I} + \mathbf{V}\mathbf{G})^{-1} \mathbf{V}\mathbf{g}_{(\omega)} \right\}$. Large values of $CD_{\omega}(\gamma)$ highlight points for special attention.

Each diagnostic tool is a function of $\mathbf{R}, \mathbf{C} = \mathbf{H}^{-1}$ and $\mathbf{e}$. These are the measures of leverages and residuals in the mixed model (Zewotir and Galpin, 2007). Thus calculations of leverages and residuals play a central role in the computation of the diagnostic tools derived in this section. In all the diagnostic tools the case deletion diagnostic measures can be found as a special case by setting ω sufficiently close to zero.

5. Illustrative example

Table 1 comprises data from a study of high-efficiency particulate air filter cartridges used in commercial respirators to prevent or reduce the respiration of toxic fumes, radionuclides, dust, mist, and other particulate matter. The aim is to determine the factors that contribute most to the variability in penetration of the filters and to determine whether the standard aerosol can be replaced by an alternative aerosol in quality assurance testing. Filters fail a quality test if the percentage of penetration of an aerosol is too large. Two aerosols were crossed with two filter manufacturers. Within each manufacturer, three

filters were used to evaluate the penetration of the two aerosols. The aerosols and manufacturers were taken as fixed effects and the filters, nested within manufacturers, were taken as a random effect. The data are from Beckman et al. (1987) who developed local influence diagnostics for the mixed model. Christensen et al. (1992), Haslett and Dillane (2004) and Zewotir and Galpin (2005) used the same data for comparison and illustration of their diagnostic methods. The same data is used here to illustrate diagnostic tools derived in this paper. All the calculations can be done with standard software; SAS macros were used here.

In our result cases 33, 31 and 32 are the most local influential points, in this order, on the variance ratio (see Figure 1). However, case 14 is not demonstrated as a local influential point. For a complete understanding of the influence of cases 33, 31, 32 and 14, it is necessary to know the behaviour of the filter variance ratio estimate graphically at different values of ω . The influence graph, Figure 2, clearly demonstrates that cases 33 and 31 are the most influential cases. We also observe that as ω goes far from the null perturbation ($\omega = 1$) and moves towards zero the influence of case 14 becomes more visible. That is why Zewotir and Galpin (2005) noted case 14 as influential with cases 32, 31 and 33 in their case deletion influential points list. It seems clear that case deletion diagnostics alone are not sufficient to handle concerns other than gross errors. It must be noted that the goal of our work was not to detect outliers but to detect cases that matter in the linear mixed model.

Cases 14 and 13 are the most locally influential observations on fixed effects estimates, random effect predictor (Figures 3 and 4). Moreover, these two cases appear as the most influential cases of the likelihood function, see Figure 5.

6. Discussion

In the past two decades, case deletion has become a popular and extremely useful basis for studying sensitivity in statistical problems. Case deletion, however, is just one possible way in which a postulated model might be perturbed meaningfully; it is worthwhile to stress parallel developments in other approaches of model perturbations.

The methodology discussed in this article is designed to allow an assessment of the sensitivity of a mixed model analysis to perturbations in one of the standard assumptions. Results from this sensitivity analysis should be treated with caution. Suppose we find that perturbation to the i^{th} error variance in the mixed model causes large changes to the parameter estimates. This finding does not tell us much about the i^{th} case of the data. It does obviously tell us that the assumption of the equality of the i^{th} term error variance to the other term variances is influential as far as estimation/prediction is concerned. It does not allow us to infer that the variance of the i^{th} case was different to the rest, for the perturbation was not being estimated, but was assumed. In other words, strongly different estimates under the perturbation suggest that the estimation procedure is highly sensitive to such perturbation and the stipulation that the i^{th} case error term has variance σ_e^2/ω rather than σ_e^2 will lead to substantial changes in the results of the analysis.

The common approach to see if perturbations of the model influence key results of analysis is an index plot. Besides ranking or index plot, for a more complete understanding of the influence of a single case, it is necessary to investigate the behaviour of the analogized Cook's distance and likelihood distance at different values of ω .

All the diagnostic measures are a function of the basic building blocks ($\mathbf{R}, \mathbf{C}, \mathbf{e}$ and $\hat{\sigma}_e^2$) but combine information differently. The basic building

blocks are computed only once from the complete data analysis. Moreover, the basic building blocks are important for identification of outlying responses and high-leverage points in the fixed and random effects design spaces. This possibility to identify specific sensitivity makes the procedures more appealing and inexpensive.

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TABLES AND FIGURES

Table 1. Per cent penetration of two aerosols using three filters from manufacturers

Aerosol 1				Aerosol 2			
Case	Manu- facturer	Filter	Penetra- tion	Case	Manu- facturer	Filter	Penetra- tion
1	1	1	0.750	19	1	1	1.120
2	1	1	0.770	20	1	1	1.100
3	1	1	0.840	21	1	1	1.120
4	1	2	0.082	22	1	2	0.160
5	1	2	0.085	23	1	2	0.110
6	1	2	0.096	24	1	2	0.260
7	1	3	0.082	25	1	3	0.150
8	1	3	0.076	26	1	3	0.120
9	1	3	0.077	27	1	3	0.120
10	2	4	0.600	28	2	4	0.910
11	2	4	0.680	29	2	4	0.830
12	2	4	0.870	30	2	4	0.950
13	2	5	4.100	31	2	5	0.660
14	2	5	5.000	32	2	5	0.830
15	2	5	1.800	33	2	5	0.610
16	2	6	1.000	34	2	6	2.170
17	2	6	1.800	35	2	6	1.520
18	2	6	2.700	36	2	6	1.580

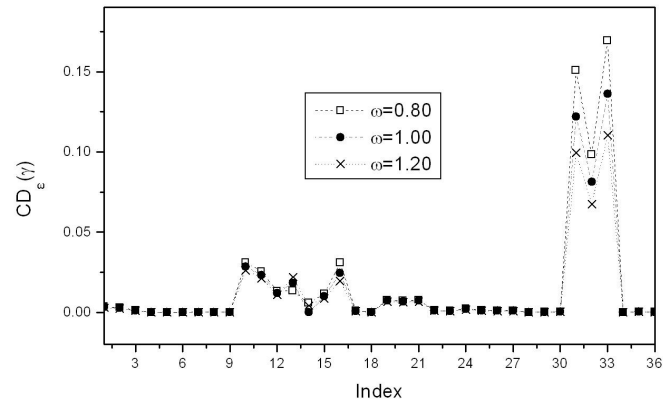


Figure 1. Index plot of $CD_{\varpi}(\gamma)$ for different values of ϖ

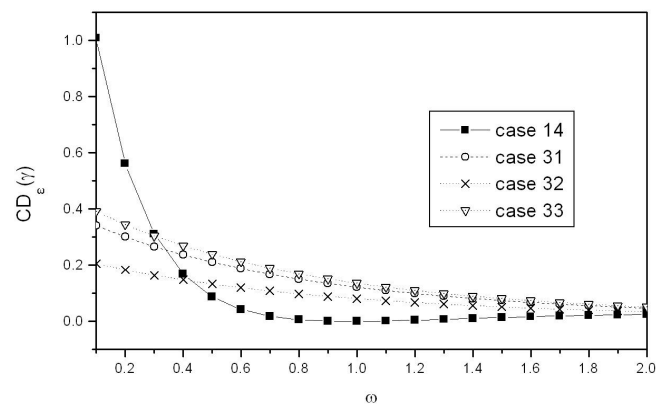


Figure 2. $CD_{\varpi}(\gamma)$ versus ϖ for cases 14, 31, 32 and 33

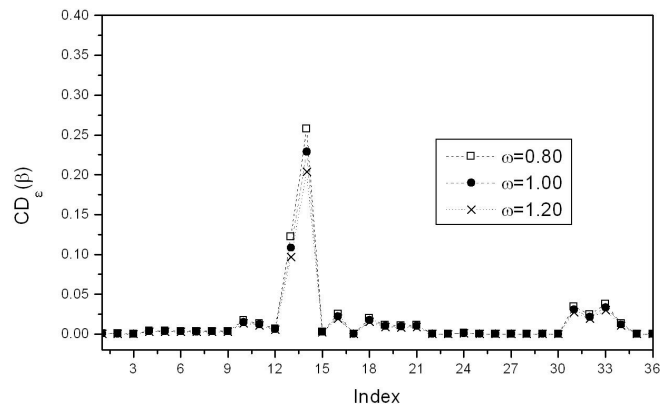


Figure 3. Index plot of $CD_{\varpi}(\beta)$ for different values of ϖ

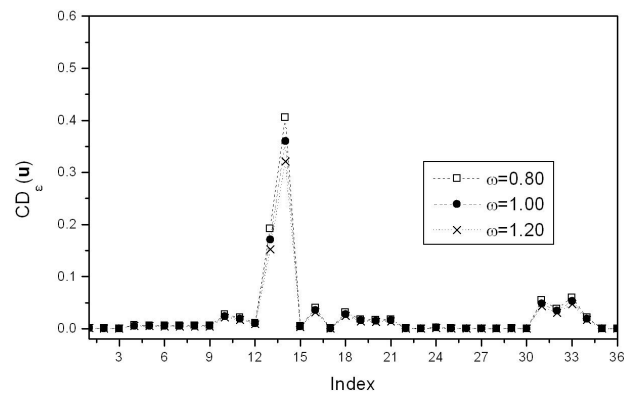


Figure 4. Index plot of $CD_{\varpi}(\mathbf{u})$ for different values of ϖ

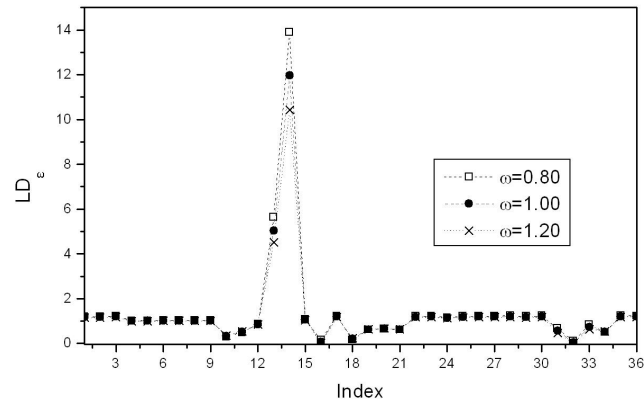


Figure 5. Index plot of LD_{ω} for different values of ω